

Classified as hazardous in accordance with the criteria of EPA New Zealand

## Section 1 - Identification

### Product Identifier

|                             |                                                                                       |
|-----------------------------|---------------------------------------------------------------------------------------|
| <b>Product Name</b>         | <u>Cyclophosphamide monohydrate</u>                                                   |
| <b>CAS No</b>               | 6055-19-2                                                                             |
| <b>Synonyms</b>             | N,N-Bis(2-chloroethyl)tetrahydro-2H-1,3,2-oxazaphosphorin-2-amine 2-oxide monohydrate |
| <b>Molecular Formula</b>    | C7 H15 Cl2 N2 O2 P . H2 O                                                             |
| <b>Molecular Weight</b>     | 279.09                                                                                |
| <b>Recommended Use</b>      | Laboratory chemicals.                                                                 |
| <b>Uses advised against</b> | No Information available                                                              |

|                                |                                                                                             |
|--------------------------------|---------------------------------------------------------------------------------------------|
| <b>Product Code</b>            | <b>L11508</b>                                                                               |
| <b>Address</b>                 | Thermo Fisher Scientific New Zealand Ltd<br>244 Bush Road, Albany,<br>Auckland, New Zealand |
| <b>Emergency Tel.</b>          | <b>CHEMTREC®</b><br><b>09 980 6780 or +64 9 980 6780</b>                                    |
| <b>Telephone / Fax Numbers</b> | Tel: 09 980 6700<br>Fax: 09 980 6788                                                        |
| <b>E-mail address</b>          | <a href="mailto:ANZinfo@thermofisher.com">ANZinfo@thermofisher.com</a>                      |

## Section 2 - Hazard(s) Identification

### Classification under Work Safe New Zealand

Classified as hazardous in accordance with the criteria of EPA New Zealand

### GHS Classification

#### Physical hazards

Based on available data, the classification criteria are not met

#### Health hazards

|                        |             |
|------------------------|-------------|
| Acute Oral Toxicity    | Category 3  |
| Germ Cell Mutagenicity | Category 1B |
| Carcinogenicity        | Category 1A |
| Reproductive Toxicity  | Category 1B |

#### Environmental hazards

Based on available data, the classification criteria are not met

### Label Elements



Signal Word

Danger

**Hazard Statements**

H301 - Toxic if swallowed

H340 - May cause genetic defects

H350 - May cause cancer

H360 - May damage fertility or the unborn child

**Precautionary Statements****Prevention**

P201 - Obtain special instructions before use

P202 - Do not handle until all safety precautions have been read and understood

P260 - Do not breathe dust/fume/gas/mist/vapors/spray

P264 - Wash face, hands and any exposed skin thoroughly after handling

P270 - Do not eat, drink or smoke when using this product

P271 - Use only outdoors or in a well-ventilated area

P280 - Wear protective gloves/protective clothing/eye protection/face protection

**Response**

P301 + P310 - IF SWALLOWED: Immediately call a POISON CENTER or doctor

P304 + P340 - IF INHALED: Remove person to fresh air and keep comfortable for breathing

P308 + P313 - IF exposed or concerned: Get medical advice/attention

P330 - Rinse mouth

**Storage**

P403 + P233 - Store in a well-ventilated place. Keep container tightly closed

P405 - Store locked up

**Disposal**

P501 - Dispose of contents/ container to an approved waste disposal plant

Other hazards which do not result in classification

## Section 3 - Composition and Information on Ingredients

| Component                    | CAS No    | Weight % |
|------------------------------|-----------|----------|
| Cyclophosphamide monohydrate | 6055-19-2 | >95      |
| Cyclophosphamide             | 50-18-0   | -        |

## Section 4 - First Aid Measures

**Description of first aid measures****General Advice**

Show this safety data sheet to the doctor in attendance. Immediate medical attention is required.

**New Zealand Emergency Tel.**CHEMTREC®  
09 980 6780 or +64 9 980 6780**Inhalation**

Remove to fresh air. If not breathing, give artificial respiration. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device.

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|                                            |                                                                                                                                                  |
|--------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
|                                            | Immediate medical attention is required.                                                                                                         |
| <b>Eye Contact</b>                         | In the case of contact with eyes, rinse immediately with plenty of water and seek medical advice.                                                |
| <b>Skin Contact</b>                        | Wash off immediately with plenty of water for at least 15 minutes. Immediate medical attention is required.                                      |
| <b>Ingestion</b>                           | Do NOT induce vomiting. Call a physician or poison control center immediately.                                                                   |
| <b>Self-Protection of the First Aider</b>  | Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination. |
| <b>First Aid Facilities</b>                | Eyewash, safety shower and washroom.                                                                                                             |
| <b>Most important symptoms and effects</b> | None reasonably foreseeable.                                                                                                                     |
| <b>Notes to Physician</b>                  | Treat symptomatically.                                                                                                                           |

## Section 5 - Fire Fighting Measures

### **Suitable Extinguishing Media**

Water spray, carbon dioxide (CO<sub>2</sub>), dry chemical, alcohol-resistant foam.

### **Extinguishing media which must not be used for safety reasons**

No information available.

### **Specific Hazards Arising from the Chemical**

Thermal decomposition can lead to release of irritating gases and vapors.

### **Hazardous Combustion Products**

Carbon monoxide (CO), Carbon dioxide (CO<sub>2</sub>), Nitrogen oxides (NO<sub>x</sub>), Oxides of phosphorus, Hydrogen chloride gas.

### **Decomposition Temperature**

90 °C

### **Special protective equipment and precautions for fire fighters**

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear. Thermal decomposition can lead to release of irritating gases and vapors.

## Section 6 - Accidental Release Measures

### **Personal Precautions, Protective Equipment and Emergency Procedures**

#### **Emergency procedures**

Ensure adequate ventilation. Use personal protective equipment as required. Avoid dust formation. Keep people away from and upwind of spill/leak. Evacuate personnel to safe areas.

#### **Environmental Precautions**

Should not be released into the environment.

#### **Methods for Containment and Clean Up**

Sweep up and shovel into suitable containers for disposal. Avoid dust formation.

#### **Precautions to prevent secondary hazards**

Clean contaminated objects and areas thoroughly observing environmental regulations

#### **Reference to Other Sections**

Refer to protective measures listed in Sections 8 and 13.

## Section 7 - Handling and Storage

### Precautions for Safe Handling

#### Advice on safe handling

Wear personal protective equipment/face protection. Do not get in eyes, on skin, or on clothing. Avoid dust formation. Use only under a chemical fume hood. Do not breathe (dust, vapor, mist, gas). Do not ingest. If swallowed then seek immediate medical assistance.

#### Hygiene Measures

Handle in accordance with good industrial hygiene and safety practice.

### Conditions for Safe Storage, Including any Incompatibilities

#### Storage Conditions

Keep containers tightly closed in a dry, cool and well-ventilated place. To maintain product quality: Keep refrigerated.

#### Incompatible Materials

Strong oxidizing agents.

AS/NZS 2243.10:2004, Safety in laboratories - Storage of chemicals

## Section 8 - Exposure Controls and Personal Protection

### Control parameters

#### Exposure limits

The product does not contain any hazardous materials with occupational exposure limits established.

#### Biological limit values

This product, as supplied, does not contain any hazardous materials with biological limits established by the region specific regulatory bodies

### Appropriate engineering controls

#### Engineering Measures

Use only under a chemical fume hood. Ensure that eyewash stations and safety showers are close to the workstation location. Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

### Individual protection measures, such as personal protective equipment

#### Eye Protection

Wear safety glasses with side shields (or goggles) (Australian/New Zealand Standard AS/NZS 1337 - Eye protectors for Industrial applications)

#### Hand Protection

Protective gloves

| Glove material                                 | Breakthrough time                 | Glove thickness | AUS/NZ Standard | Glove comments        |
|------------------------------------------------|-----------------------------------|-----------------|-----------------|-----------------------|
| Natural rubber, Nitrile rubber, Neoprene, PVC. | See manufacturers recommendations | -               | AS/NZS 2161     | (minimum requirement) |

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatibility, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

#### Skin and body protection

Long sleeved clothing

|                                        |                                                                                                                                                                                                                                                                                                                             |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Respiratory Protection</b>          | Use an AS/NZS 1716 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced. To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained in line with AS/NZS 1715 on the use and maintenance of respiratory protective devices |
| <b>Recommended Filter type:</b>        | Particulates filter conforming to EN 143 (or AUS/NZ equivalent)                                                                                                                                                                                                                                                             |
| <b>Recommended half mask:-</b>         | Particle filtering: EN149:2001 (or AUS/NZ equivalent)<br>When RPE is used a face piece Fit Test should be conducted                                                                                                                                                                                                         |
| <b>Hygiene Measures</b>                | Handle in accordance with good industrial hygiene and safety practice.                                                                                                                                                                                                                                                      |
| <b>Environmental exposure controls</b> | No information available.                                                                                                                                                                                                                                                                                                   |

## Section 9 - Physical and Chemical Properties

### Information on basic physical and chemical properties

|                                                |                               |                                          |
|------------------------------------------------|-------------------------------|------------------------------------------|
| <b>Physical State</b>                          | Solid                         |                                          |
| <b>Appearance</b>                              | Off-white                     |                                          |
| <b>Odor</b>                                    | Odorless                      |                                          |
| <b>Odor Threshold</b>                          | No data available             |                                          |
| <b>pH</b>                                      | 4.0-6.0                       | 2% aq.sol                                |
| <b>Melting Point/Range</b>                     | 49 - 53 °C / 120.2 - 127.4 °F |                                          |
| <b>Softening Point</b>                         | No data available             |                                          |
| <b>Boiling Point/Range</b>                     | No information available      |                                          |
| <b>Flammability (liquid)</b>                   | Not applicable                | Solid                                    |
| <b>Flammability (solid,gas)</b>                | No information available      |                                          |
| <b>Explosion Limits</b>                        | No data available             |                                          |
| <b>Flash Point</b>                             | > 110 °C / > 230 °F           | <b>Method -</b> No information available |
| <b>Autoignition Temperature</b>                | No data available             |                                          |
| <b>Decomposition Temperature</b>               | 90 °C                         |                                          |
| <b>Viscosity</b>                               | Not applicable                | Solid                                    |
| <b>Water Solubility</b>                        | 40 g/l                        |                                          |
| <b>Solubility in other solvents</b>            | No information available      |                                          |
| <b>Partition Coefficient (n-octanol/water)</b> |                               |                                          |
| <b>Vapor Pressure</b>                          | No data available             |                                          |
| <b>Density / Specific Gravity</b>              | No data available             |                                          |
| <b>Bulk Density</b>                            | No data available             |                                          |
| <b>Vapor Density</b>                           | Not applicable                | Solid                                    |
| <b>Particle characteristics</b>                | No data available             |                                          |

### Other information

|                          |                           |
|--------------------------|---------------------------|
| <b>Molecular Formula</b> | C7 H15 Cl2 N2 O2 P . H2 O |
| <b>Molecular Weight</b>  | 279.09                    |
| <b>Evaporation Rate</b>  | Not applicable - Solid    |

## Section 10 - Stability and Reactivity

|                                         |                                            |
|-----------------------------------------|--------------------------------------------|
| <b>Reactivity</b>                       | None known, based on information available |
| <b>Stability</b>                        | Stable under normal conditions.            |
| <b>Sensitivity to Mechanical Impact</b> | No information available                   |
| <b>Sensitivity to Static Discharge</b>  | No information available                   |
| <b>Hazardous Polymerization</b>         | Hazardous polymerization does not occur.   |

**Hazardous Reactions** None under normal processing.

**Conditions to Avoid** Incompatible products, Excess heat.

**Incompatible Materials** Strong oxidizing agents.

**Hazardous Decomposition Products** Carbon monoxide (CO). Carbon dioxide (CO<sub>2</sub>). Nitrogen oxides (NO<sub>x</sub>). Oxides of phosphorus. Hydrogen chloride gas.

## Section 11 - Toxicological Information

### Acute Effects

### Information on likely routes of exposure

#### Product Information

**Inhalation** Irritating to respiratory system. May be harmful if inhaled.  
**Eyes** Irritating to eyes.  
**Skin** Irritating to skin. May be harmful in contact with skin.  
**Ingestion** Toxic if swallowed. May cause adverse liver effects. May cause adverse kidney effects. Ingestion may cause gastrointestinal irritation, nausea, vomiting and diarrhea.

### Numerical measures of toxicity

#### (a) acute toxicity;

**Oral** Category 3  
**Dermal** No data available  
**Inhalation** No data available

| Component                    | LD50 Oral                | LD50 Dermal | LC50 Inhalation |
|------------------------------|--------------------------|-------------|-----------------|
| Cyclophosphamide monohydrate | LD50 = 94 mg/kg ( Rat )  |             |                 |
| Cyclophosphamide             | LD50 = 100 mg/kg ( Rat ) |             |                 |

**(b) skin corrosion/irritation;** No data available

**(c) serious eye damage/irritation;** No data available

#### (d) respiratory or skin sensitization;

**Respiratory** No data available  
**Skin** No data available

**(e) germ cell mutagenicity;** Category 1B

May cause heritable genetic damage

**(f) carcinogenicity;** Category 1A

Possible cancer hazard. May cause cancer based on animal data The table below indicates whether each agency has listed any ingredient as a carcinogen

| Component                    | New Zealand | Australia             | New South Wales | Western Australia     | IARC    | EU | UK | Germany |
|------------------------------|-------------|-----------------------|-----------------|-----------------------|---------|----|----|---------|
| Cyclophosphamide monohydrate |             |                       |                 |                       | Group 1 |    |    |         |
| Cyclophosphamide             |             | Notifiable Carcinogen | Notifiable      | Requires Approval for | Group 1 |    |    |         |

|  |  |  |  |     |  |  |  |  |
|--|--|--|--|-----|--|--|--|--|
|  |  |  |  | Use |  |  |  |  |
|--|--|--|--|-----|--|--|--|--|

(g) reproductive toxicity;  
Reproductive Effects  
Developmental Effects

Category 1B  
Experiments have shown reproductive toxicity effects on laboratory animals  
May cause harm to the unborn child

(h) STOT-single exposure; No data available

(i) STOT-repeated exposure; No data available  
Target Organs No information available.

(j) aspiration hazard; Not applicable  
Solid

Other Adverse Effects The toxicological properties have not been fully investigated.

Symptoms / effects, both acute and delayed  
No information available.

## Section 12 - Ecological Information

### Ecotoxicity

Aquatic ecotoxicity Do not empty into drains.

Terrestrial ecotoxicity There is no data for this product

### Persistence and Degradability

Persistence Soluble in water, Persistence is unlikely, based on information available.

Bioaccumulative Potential Bioaccumulation is unlikely

Mobility The product is water soluble, and may spread in water systems. Will likely be mobile in the environment due to its water solubility. Highly mobile in soils

### Other adverse effects

### Endocrine Disruptor Information

| Component        | EU - Endocrine Disruptors<br>Candidate List | EU - Endocrine Disruptors -<br>Evaluated Substances | Japan - Endocrine Disruptor<br>Information |
|------------------|---------------------------------------------|-----------------------------------------------------|--------------------------------------------|
| Cyclophosphamide | Group III Chemical                          |                                                     |                                            |

Persistent Organic Pollutant This product does not contain any known or suspected substance

Ozone Depletion Potential This product does not contain any known or suspected substance

## Section 13 - Disposal Considerations

### Waste treatment methods

Waste from Residues/Unused Do not allow into drains or watercourses or dispose of where ground or surface waters may

|                               |                                                                                                                                                                                                                                                     |
|-------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Products</b>               | be affected. Wastes, including emptied containers, are controlled wastes and should be disposed of in accordance with all federal, E.P.A., state and local regulations. Assure conformity with all applicable regulations.                          |
| <b>Contaminated Packaging</b> | Dispose of this container to hazardous or special waste collection point.                                                                                                                                                                           |
| <b>Other Information</b>      | Disposal agencies or waste contractors must comply with the New Zealand Hazardous Substances (Disposal) Regulations . Waste codes should be assigned by the user based on the application for which the product was used. Do not empty into drains. |

## Section 14 - Transport Information

### NZS 5433:2020

|                                |                                                 |
|--------------------------------|-------------------------------------------------|
| <b>UN-No</b>                   | UN3464                                          |
| <b>Proper Shipping Name</b>    | Organophosphorus compound, toxic, solid, n.o.s. |
| <b>Technical Shipping Name</b> | Cyclophosphamide monohydrate ,Cyclophosphamide  |
| <b>Hazard Class</b>            | 6.1                                             |
| <b>Packing Group</b>           | III                                             |

### IATA

|                                |                                                 |
|--------------------------------|-------------------------------------------------|
| <b>UN-No</b>                   | UN3464                                          |
| <b>Proper Shipping Name</b>    | Organophosphorus compound, toxic, solid, n.o.s. |
| <b>Technical Shipping Name</b> | Cyclophosphamide monohydrate ,Cyclophosphamide  |
| <b>Hazard Class</b>            | 6.1                                             |
| <b>Packing Group</b>           | III                                             |

### IMDG/IMO

|                                |                                                 |
|--------------------------------|-------------------------------------------------|
| <b>UN-No</b>                   | UN3464                                          |
| <b>Proper Shipping Name</b>    | Organophosphorus Compound, toxic, solid, n.o.s. |
| <b>Technical Shipping Name</b> | Cyclophosphamide monohydrate ,Cyclophosphamide  |
| <b>Hazard Class</b>            | 6.1                                             |
| <b>Packing Group</b>           | III                                             |

|                              |                       |
|------------------------------|-----------------------|
| <b>Environmental hazards</b> | No hazards identified |
|------------------------------|-----------------------|

|                                                                                 |                                |
|---------------------------------------------------------------------------------|--------------------------------|
| <b>Transport in bulk according to Annex II of MARPOL 73/78 and the IBC Code</b> | Not applicable, packaged goods |
|---------------------------------------------------------------------------------|--------------------------------|

|                            |                                                                                                                         |
|----------------------------|-------------------------------------------------------------------------------------------------------------------------|
| <b>Special Precautions</b> | No special precautions required. Please refer to the applicable dangerous goods regulations for additional information. |
|----------------------------|-------------------------------------------------------------------------------------------------------------------------|

|                               |            |
|-------------------------------|------------|
| <b>Additional information</b> | None known |
|-------------------------------|------------|

## Section 15 - Regulatory Information

### Safety, health and environmental regulations/legislation specific for the substance or mixture

#### **National Regulations**

There are no applicable tolerable exposure limits or environmental exposure limits according to the EPA Controls for Hazardous Substances

#### **Certified handlers, tracking and controlled substance license requirements**

Certified handlers are required for some substances. This includes substances requiring a controlled substance license, and most explosives, vertebrates toxic agents, and certain fumigants. Acutely toxic substances which are a Category 1 or 2, such as pesticides also require Certified handlers. Please check the Health and Safety at Work Act 2015 for further information. Tracking is required for some highly hazardous substances. These substances need to be under the control of an appropriately trained person



or appropriately secured. Please check the Health and Safety at Work Act 2015 for further information.

#### Prohibition or notification/licensing requirements

Shown below are details of specific prohibition/notifications or licensing requirements when they apply.

#### International Regulations

|                                     |                                                                |
|-------------------------------------|----------------------------------------------------------------|
| <b>Ozone Depletion Potential</b>    | This product does not contain any known or suspected substance |
| <b>Persistent Organic Pollutant</b> | This product does not contain any known or suspected substance |
| <b>Rotterdam Convention (PIC)</b>   | Not applicable                                                 |

#### Authorisation/Restrictions according to EU REACH

| Component        | REACH (1907/2006) - Annex XIV - Substances Subject to Authorization | REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances | REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC) |
|------------------|---------------------------------------------------------------------|-------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Cyclophosphamide | -                                                                   | Use restricted. See item 75. (see link for restriction details)               | -                                                                                                     |

<https://echa.europa.eu/substances-restricted-under-reach>

#### International Inventories

New Zealand (NZIoC), Australia (AICS), Europe (EINECS/ELINCS/NLP), Korea (KECL), China (IECSC), Taiwan (TCSI), Japan (ENCS), Japan (ISHL), Canada (DSL/NDSL), Philippines (PICCS). US EPA (TSCA) - Toxic Substances Control Act, (40 CFR Part 710)

| Component                    | CAS No    | NZIoC | AICS | EINECS | ELINCS | NLP | KECL | IECSC | TCSI |
|------------------------------|-----------|-------|------|--------|--------|-----|------|-------|------|
| Cyclophosphamide monohydrate | 6055-19-2 | X     | -    | -      | -      | -   | -    | -     | X    |
| Cyclophosphamide             | 50-18-0   | -     | -    | -      | -      | -   | -    | -     | -    |

| Component                    | CAS No    | TSCA | TSCA Inventory notification - Active-Inactive | DSL | NDSL | PICCS | ISHL | ENCS |
|------------------------------|-----------|------|-----------------------------------------------|-----|------|-------|------|------|
| Cyclophosphamide monohydrate | 6055-19-2 | -    | -                                             | -   | -    | -     | -    | -    |
| Cyclophosphamide             | 50-18-0   | -    | -                                             | X   | -    | -     | -    | X    |

**Legend:** X - Listed '-' - Not Listed

**KECL** - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

## Section 16 - Other Information

**This safety data sheet complies with the requirements of the EPA Hazardous Substances (Hazard Classification) Notice 2020 and WorkSafe New Zealand Regulations**

#### Legend

**NZIoC** - New Zealand Inventory of Chemicals  
**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory  
**DSL/NDSL** - Canadian Domestic Substances List/Non-Domestic Substances List  
**IECSC** - Chinese Inventory of Existing Chemical Substances  
**PICCS** - Philippines Inventory of Chemicals and Chemical Substances  
**TWA** - Time Weighted Average  
**IARC** - International Agency for Research on Cancer  
**NZS 5433:2020** - Transport of Dangerous Goods on Land  
**ICAO/IATA** - International Civil Aviation Organization/International Air

**AICS** - Australian Inventory of Chemical Substances  
**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances  
**ENCS** - Japanese Existing and New Chemical Substances  
**KECL** - Korean Existing and Evaluated Chemical Substances  
**CAS** - Chemical Abstracts Service  
**ACGIH** - American Conference of Governmental Industrial Hygienists  
**PNEC** - Predicted No Effect Concentration  
**OECD** - Organisation for Economic Co-operation and Development  
**IMO/IMDG** - International Maritime Organization/International Maritime

**MARPOL** - International Convention for the Prevention of Pollution from Ships

**LD50** - Lethal Dose 50%

**EC50** - Effective Concentration 50%

**WEL** - Workplace Exposure Limit

**DNEL** - Derived No Effect Level

**POW** - Partition coefficient Octanol:Water

**vPvB** - very Persistent, very Bioaccumulative

**VOC** - (Volatile Organic Compound)

Dangerous Goods Code

**ADG** - Australian Code for the Transport of Dangerous Goods by Road and Rail

**LC50** - Lethal Concentration 50%

**ATE** - Acute Toxicity Estimate

**RPE** - Respiratory Protective Equipment

**NOEC** - No Observed Effect Concentration

**BCF** - Bioconcentration factor

**PBT** - Persistent, Bioaccumulative, Toxic

#### Key literature references and sources for data

HSNO classifications provided in the New Zealand Chemical Classification Information Database (CCID).

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

EPA Guide to classifying hazardous substances in New Zealand

EPA - Assigning a product to an existing HSNO approval guide

#### Training Advice

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.

Use of personal protective equipment, covering appropriate selection, compatibility, breakthrough thresholds, care, maintenance, fit and standards.

First aid for chemical exposure, including the use of eye wash and safety showers.

|                         |                |
|-------------------------|----------------|
| <b>Revision Date</b>    | 15-Mar-2023    |
| <b>Revision Summary</b> | Not applicable |

#### Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

## End of Safety Data Sheet